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ACX7100/ACX7509 Validation: VPLS

An HPE Juniper Networking Techpost

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1 VPLS Validation on ACX7000

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Let's verify we can support 8,000 VPLS instances in the ACX7000 products with 640,000 MAC addresses.

1.1 Introduction

This is the fifth article in the ACX7k Metro Validation Series:

- 1. EVPN MAC-VRF (E-LAN)
- 2. EVPN VPWS
- 3. L3VPN, 6VPE
- 4. L2VPN
- 5. VPLS
- 6. L2 MAC Scale and Learning Rate

In this article, we validate the VPLS feature scale support on the ACX7100-32C platform with 22.2R1 Junos-EVO build.

VPLS is a BGP or LDP MPLS Enabled Solution to connect Customer Edge Devices (CEs) across a Provider Edge Network. RFC4761 describes VPLS using BGP for Auto-Discovery and Signalling and RFC4762 presents VPLS using LDP Signalling. The PEs provide Layer 2 bridged connectivity between CEs. The PEs of the network are connected via MPLS technology with the help of LDP, RSVP, SR-MPLS, etc. In VPLS, the MAC address associated with the hosts is learned via the data plane.

JUNOS-EVO supports VPLS functionality with the instance type "virtual-switch" on ACX7000 platforms. In this article, we define the support and scale of the VPLS on the ACX7100-32C platform with LDP Signalling.

ACX7100-32C with 22.2R1 Junos Version supports 8,000 VPLS instances. The Key Performance Indicator (KPI) tested are captured in the below table. The same scale is tested on ACX7100-48L as well as on ACX7509 Scale is not covered in this article and supports a different scale.

> Please note that despite sharing the same ACX moniker, the ACX7000 products are different products than ACX500/710/1000/1100/2100/2200/4000/5000/5400/6000. They are powered by different Packet Forwarding Engines (PFE), and support different feature sets and scales.

The Key Performance Indicator (KPI) tested are:

KPI	Scale
VPLS Instance	8,000
Number of VLANs	8,000
Number of interfaces	8,000
Number of MAC addresses	640,000

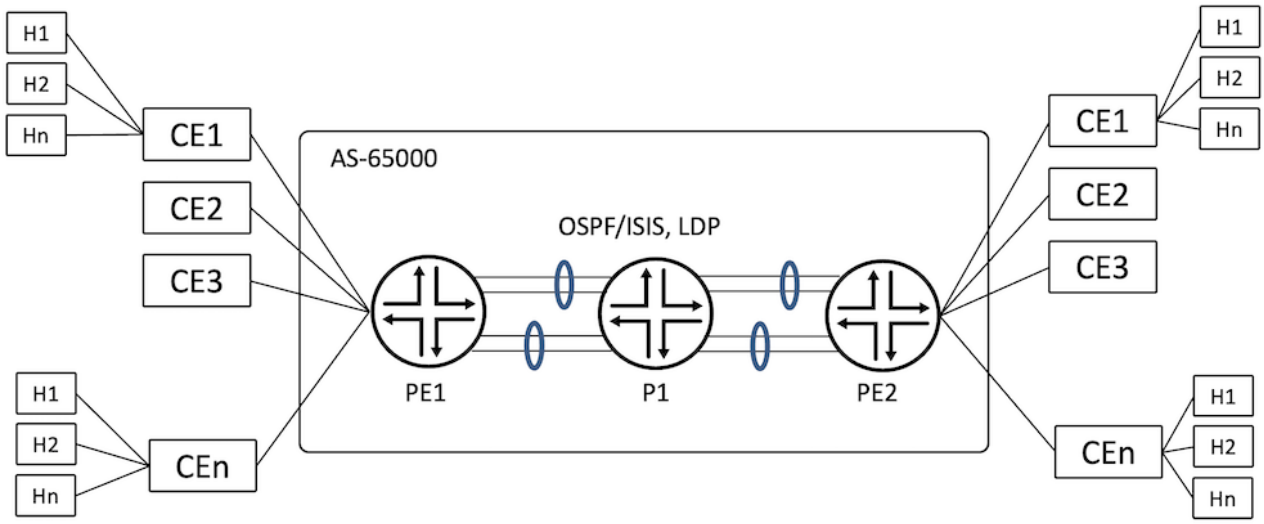
1.2 Metro VPLS Topology

Test topology consists of three routers: PE1, P1 and PE2, where :

- ACX7100-32C is PE1
- ACX7100-48L is PE2
- P1 is a PTX10008

PEs are connected to CEs using 4x 100GE links. CEs are simulated using a traffic generator. PEs are connected to the core using 8x 100GE links, which are bundled in 2x LAGs each having four links.

The ACX7100-32C is the Device Under Test (DUT). The underlay transport used for testing is LDP with both OSPF and ISIS protocols individually. The CE-bound interface configurations are of SP-Style (see glossary).



Test configures 8,000 virtual-switch instances, 80 MACs are learned per instance, 40 from local and 40 from remote peers. Bidirectional traffic in iMIX mode at 99.9% offer-load flows for all the VPLS services. A total of ~800Gbps traffic transits through the DUT.

1.3 VPLS Configurations

This section covers the configurations required to bring up VPLS using LDP signalling. In the PE Network, we can have OSPF or ISIS as the IGP protocol for the node connectivity and LDP or RSVP helps to bring up the transport protocol. BGP Auto-Discovery and ldp-signalling is enabled across the PE devices to enable VPLS services.

1.3.1 Configurations at PE1

```
regress@PE1> show configuration protocols |display inheritance no-comments

bgp {
   group IBGP {
      type internal;
      local-address 12.1.1.1;
      family inet {
         unicast;
      }
      family l2vpn {
         signaling;
      }
```

```
        neighbor 12.1.1.3 {
            export nhself;
        }
    }
}
isis {
    interface lo0.0 {
        passive;
    }
    interface ae0.0 {
        level 1 disable;
        point-to-point;
    }
    interface ae1.0 {
        level 1 disable;
        point-to-point;
    }
    level 1 disable;
    level 2 wide-metrics-only;
}
ldp {
    interface ae0.0;
    interface ae1.0;
    interface lo0.0;
}
mpls {
    icmp-tunneling;
    ipv6-tunneling;
    interface ae0.0;
    interface ae1.0;
}

regress@PE1>
```

Routing-Instance configuration required to bringup the LDP signalling based VPLS instances:

```
interfaces {
    et-0/0/0 {
        flexible-vlan-tagging;
        encapsulation flexible-ethernet-services;
        unit 1 {
            encapsulation vlan-vpls;
            vlan-id 1;
        }
    }
}
routing-instances {
    METRO_VPLS_VRF_1 {
        instance-type virtual-switch;
        protocols {
            vpls {
                neighbor 12.1.1.3;
                site-range 65534;
                label-block-size 8;
                no-tunnel-services;
                vpls-id 1;
            }
        }
        switch-options {
            mac-table-size {
                5120;
            }
        }
        route-distinguisher 12.1.1.1:1;
        vrf-target target:65000:1;
        vlans {
            VLAN_1 {
                vlan-id 1;
                interface et-0/0/0.1;
            }
        }
    }
}
```

1.3.2 CE Bound Interface Schema

The PE devices are connected to the traffic generator using 4x 100GE interfaces, each interface is logically split into sub-interfaces using VLANs, one interface bound to one VPLS instance.

PE1	PE2	Port	Service Type	VRF	VLAN Association	Number of MAC	Traffic Load
et-0/0/0	et-0/0/48:0	P1	virtual-switch	1-2000	1-2000	160000	99.9%
et-0/0/1	et-0/0/48:1	P2	virtual-switch	2001-4000	1-2000	160000	99.9%
et-0/0/2	et-0/0/48:2	P3	virtual-switch	4001-6000	1-2000	160000	99.9%
et-0/0/3	et-0/0/48:3	P4	virtual-switch	6001-8000	1-2000	160000	99.9%

1.4 VPLS Verification

The first step to verify the VPLS service is to make sure that PE devices are having BGP established with l2vpn-signalling. As shown below from PE1, BGP is in established state for the Peer PE2 (12.1.1.3):

```
regress@PE1> show bgp summary
Threading mode: BGP I/O
Default eBGP mode: advertise - accept, receive - accept
Groups: 1 Peers: 1 Down peers: 0
Table Tot Paths Act Paths Suppressed History Damp State Pending
inet.0
                   9 1 0 0 0 0
inet6.0
                   0 0 0 0 0 0
bgp.l2vpn.0
                   0 0 0 0 0 0
inet6.3
                   0 0 0 0 0 0
Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
12.1.1.3 65000 7 6 0 0 10 Establ
  inet.0: 1/9/9/0
  bgp.l2vpn.0: 0/0/0/0
  METRO_VPLS_VRF_1.l2vpn.0: 0/0/0/0
  METRO_VPLS_VRF_2.l2vpn.0: 0/0/0/0
  METRO_VPLS_VRF_3.l2vpn.0: 0/0/0/0
...
  METRO_VPLS_VRF_7998.l2vpn.0: 0/0/0/0
  METRO_VPLS_VRF_7999.l2vpn.0: 0/0/0/0
  METRO_VPLS_VRF_8000.l2vpn.0: 0/0/0/0

regress@PE1>
```

The MAC-Table Summary captures 640,000 MACs of the host.

```
regress@PE1> show ethernet-switching table summary
Total dynamic and static MAC addresses learned globally : 640000
Configured static MAC addresses learned globally : 0
```

The VPLS connections is UP towards the 12.1.1.3 Peer for the routing-instance METRO_VPLS_VRF_1

```
regress@PE1> show vpls connections instance METRO_VPLS_VRF_1
Layer-2 VPN connections:

```

```
Legend for connection status (St)
EI -- encapsulation invalid NC -- interface encapsulation not CCC/TCC/VPLS
EM -- encapsulation mismatch WE -- interface and instance encaps not same
VC-Dn -- Virtual circuit down NP -- interface hardware not present
CM -- control-word mismatch -> -- only outbound connection is up
CN -- circuit not provisioned <- -- only inbound connection is up
OR -- out of range Up -- operational
OL -- no outgoing label Dn -- down
LD -- local site signaled down CF -- call admission control failure
RD -- remote site signaled down SC -- local and remote site ID collision
LN -- local site not designated LM -- local site ID not minimum designated
RN -- remote site not designated RM -- remote site ID not minimum designated
XX -- unknown connection status IL -- no incoming label
MM -- MTU mismatch MI -- Mesh-Group ID not available
BK -- Backup connection ST -- Standby connection
PF -- Profile parse failure PB -- Profile busy
RS -- remote site standby SN -- Static Neighbor
LB -- Local site not best-site RB -- Remote site not best-site
VM -- VLAN ID mismatch HS -- Hot-standby Connection

Legend for interface status
Up -- operational
Dn -- down

Instance: METRO_VPLS_VRF_1
Edge protection: Not-Primary
  LDP-VPLS State
  VPLS-id: 1
  Mesh-group connections: __ves__
    Neighbor Type St Time last up # Up trans
    12.1.1.3(vpls-id 1) rmt Up Oct 27 09:49:15 2022 1
      Remote PE: 12.1.1.3, Negotiated control-word: No
      Incoming label: 16, Outgoing label: 16
      Negotiated PW status TLV: No
      Local interface: lsi.1048576, Status: Up, Encapsulation: ETHERNET
        Description: Intf - vpls METRO_VPLS_VRF_1 neighbor 12.1.1.3 vpls-id 1
      Flow Label Transmit: No, Flow Label Receive: No

regress@PE1>
```

The VLAN Association of the METRO_MAC_VRF_1 instance is shown below.

```
regress@PE1> show vlans VLAN_1

Routing instance VLAN name Tag Interfaces
METRO_VPLS_VRF_1 VLAN_1 1
                                            et-0/0/0.1*
                                            lsi.1048576*

regress@PE1>
```

The VPLS Forwarding Table of METRO_VPLS_VRF_1 instance captures the MAC addresses from both local and remote interfaces.

```
regress@PE1> show ethernet-switching table instance METRO_VPLS_VRF_1

MAC flags (S - static MAC, D - dynamic MAC, L - locally learned, P - Persistent static, C -
    Control MAC
        SE - statistics enabled, NM - non configured MAC, R - remote PE MAC, O - ovsdb MAC)

Ethernet switching table : 80 entries, 80 learned
Routing instance : METRO_VPLS_VRF_1
   Vlan MAC MAC Age Logical NH RTR
   name address flags interface Index ID
   VLAN_1 00:11:01:00:00:01 D - et-0/0/0.1 0 0
   VLAN_1 00:11:01:00:00:02 D - et-0/0/0.1 0 0
...
   VLAN_1 00:11:01:00:00:27 D - et-0/0/0.1 0 0
   VLAN_1 00:11:01:00:00:28 D - et-0/0/0.1 0 0
   VLAN_1 00:15:01:00:00:01 D - lsi.1048576 0 0
   VLAN_1 00:15:01:00:00:02 D - lsi.1048576 0 0
....
   VLAN_1 00:15:01:00:00:27 D - lsi.1048576 0 0
   VLAN_1 00:15:01:00:00:28 D - lsi.1048576 0 0

regress@PE1>
```

The following output captures the total number of instances and total MAC learned in the DUT.

```
regress@PE1> show vpls connections |match "rmt Up" | count
Count: 8000 lines

regress@PE1> show ethernet-switching table summary
Total dynamic and static MAC addresses learned globally : 640000
Configured static MAC addresses learned globally : 0

regress@PE1> show route summary
Autonomous system number: 65000
Router ID: 12.1.1.1

Highwater Mark (All time / Time averaged watermark)
   RIB unique destination routes: 32046 at 2022-10-27 09:49:42 / 0
   RIB routes : 32054 at 2022-10-27 09:49:42 / 0
   FIB routes : 16032 at 2022-10-27 09:49:42 / 0
   VRF type routing instances : 0 at 2022-10-27 09:43:24

inet.0: 28 destinations, 36 routes (28 active, 0 holddown, 0 hidden)
           Direct: 4 routes, 4 active
            Local: 3 routes, 3 active
              BGP: 9 routes, 1 active
           Static: 14 routes, 14 active
            IS-IS: 5 routes, 5 active
              LDP: 1 routes, 1 active

inet.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
              LDP: 2 routes, 2 active

iso.0: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
           Direct: 1 routes, 1 active

mpls.0: 16009 destinations, 16009 routes (16009 active, 0 holddown, 0 hidden)
            MPLS: 6 routes, 6 active
             LDP: 3 routes, 3 active
            VPLS: 16000 routes, 16000 active

inet6.0: 4 destinations, 4 routes (4 active, 0 holddown, 0 hidden)
           Direct: 2 routes, 2 active
              BGP: 1 routes, 1 active
           INET6: 1 routes, 1 active

inet6.3: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
              LDP: 2 routes, 2 active

l2circuit.0: 16000 destinations, 16000 routes (16000 active, 0 holddown, 0 hidden)
             LDP: 8000 routes, 8000 active
            VPLS: 8000 routes, 8000 active

regress@PE1>
```

1.4.1 Traffic Verification at Router at PE1

Traffic load-balanced across the LAG bundles ae0 and ae1 towards the core and among the member links of the LAG's.

```
PE1 Seconds: 8 Time: 20:19:40

Interface Link Input packets (pps) Output packets (pps)

 ae0 Up 3868838703698 (31186310) 3880085080678 (31271792)
 ae1 Up 3890226133290 (31360142) 3879121784169 (31272766)
 esi Up 0 (0) 0 (0)
 et-0/0/0 Up 1939850875296 (15635490) 1939627063135 (15635223)
 et-0/0/1 Up 1939850824172 (15635475) 1939625876351 (15635416)
 et-0/0/2 Up 1939850765486 (15635220) 1939626354193 (15635382)
 et-0/0/3 Up 1939850784451 (15635227) 1939627926794 (15635176)
 et-0/0/4 Up 0 (0) 4478 (0)
 et-0/0/5 Down 0 (0) 0 (0)
 et-0/0/6 Up 964863965085 (7776937) 970021565092 (7817826)
 et-0/0/7 Up 963482838231 (7766530) 970046108276 (7818076)
 et-0/0/8 Up 973253565004 (7845764) 969996802574 (7817824)
 et-0/0/9 Up 967238335378 (7797079) 970020604736 (7818066)
 et-0/0/10 Up 965735799928 (7784996) 969828700176 (7818528)
```

```
et-0/0/11 Up 980547733423 (7904378) 969732569828 (7817790)
et-0/0/12 Up 970951694823 (7827028) 969804548668 (7818413)
et-0/0/13 Up 972990905116 (7843740) 969755965497 (7818035)
```

1.5 Generating Scale Configuration

The Python Script is having three components, a Jinja File capturing the various configuration templates, a Params File capturing the various user variables and the script uses both these input files and generates the required configurations and commit that in the router.

Note: The configs are generated as per the CE-bound interface schema.

1.5.1 Jinja Configuration Template

The template contains configurations for both interfaces which are part of the routing instances and routing instance configurations.

```
# Routing Instnace Configuration
# File Name: vpls_jinja.j2
# Version: 1.0
groups {{group_name}} {
   interfaces {
      {{ifd_name}} {
         flexible-vlan-tagging;
         encapsulation flexible-ethernet-services;
         {%- for unit_id, vlan_id in ifl_variables %}
         unit {{unit_id}} {
            vlan-id {{vlan_id}};
            encapsulation vlan-vpls;
         }
         {%- endfor %}
      }
   }
   routing-instances {
      {%- for vrf_id, ifl_name, vlan_id, vpls_id in vpls_variables %}
      METRO_VPLS_VRF_{{vrf_id}} {
         instance-type virtual-switch;
         switch-options {
            {%- if mac_table_size_defined %}
            mac-table-size {
               {{mac_table_size}};
            }
            {%- endif %}
         }
         vlans {
            VLAN_{{vrf_id}} {
               vlan-id {{vlan_id}};
               interface {{ifl_name}};
            }
         }
         route-distinguisher {{router_id}}:{{vrf_id}};
         vrf-target target:{{local_as_no}}:{{vrf_id}};
         protocols {
            vpls {
               neighbor {{neighbor_id}};
               site-range {{site_range}};
               label-block-size {{label_block_size}};
               {%- if tunnel_disable %}
               no-tunnel-services;
               {%- endif %}
               vpls-id {{vpls_id}};
            }
         }
      }
      {%- endfor %}
   }
}
```

1.5.2 Configuration Parameters

‘‘‘



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